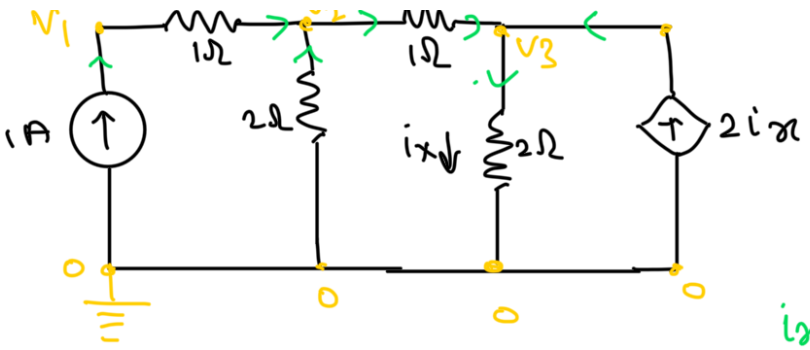



$$\frac{V_1-V_2}{1}=1$$

$$\frac{V_1-V_2}{1}+\frac{V_2}{2}=\frac{V_3-V_2}{1}$$

$$\frac{V_2-V_3}{1}=\frac{V_3}{2}+2i_x$$

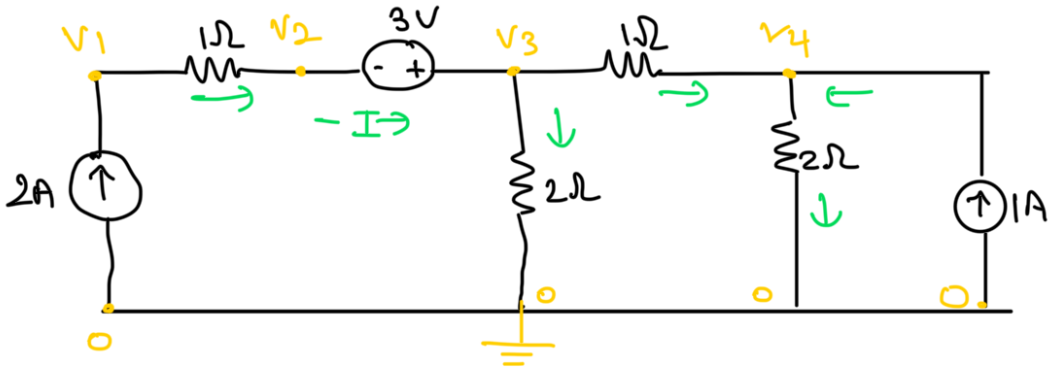
$$i_x=\frac{V_2}{2}$$




$$-3+i_1+2(i_1-i_2)=0$$

$$2(i_2-i_1)+i_2+2i_2+2i_4=0$$

$$i_4=i_1-i_2$$




$$\frac{V_1-V_2}{1}=2$$

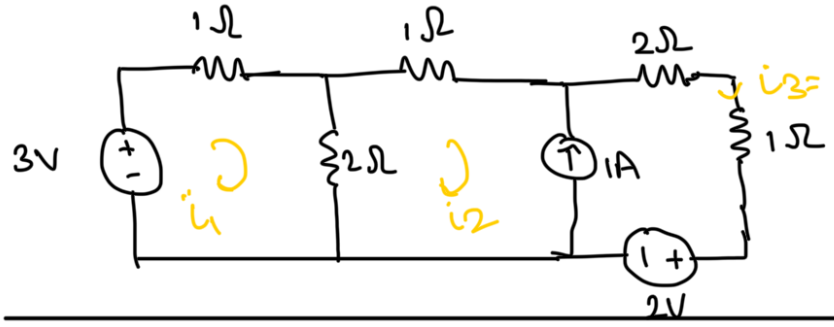
$$\frac{V_1-V_2}{1}=I$$

$$I=\frac{V_3}{2}+\frac{V_3-V_4}{1}$$

$$\frac{V_3-V_4}{1}+1=\frac{V_4}{2}$$

$$V_3=V_2+3$$

(we can also combine the current and voltage source in series and neglect the voltage source)




$$i_2\cdot 1=i_3\cdot 2+i_3\cdot 1+2$$

$$-3+i_4\cdot 1+(i_4-i_2)\cdot 2=0$$

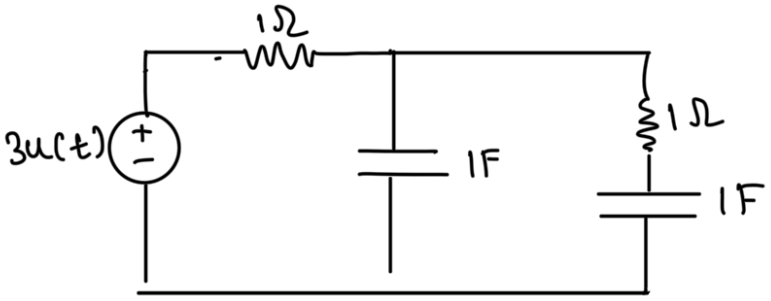
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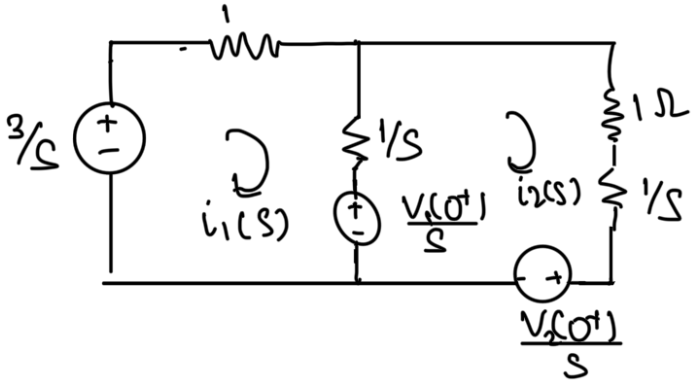
Approach:

Count mesh, nodes, current sources, voltage sources.

→ Nodal: effective equations = # nodes − # V.S

→ Mesh: effective equations = # mesh − # C.S




## Network Theorems:

**Linear Network:** R/C/L + independent sources + dependent linear sources (diodes and don't come here)

**Substitution Theorm:** Replace a resistor by a voltage/current source with the same voltage drop / current through the resistor ⇒ NOTHING CHANGES

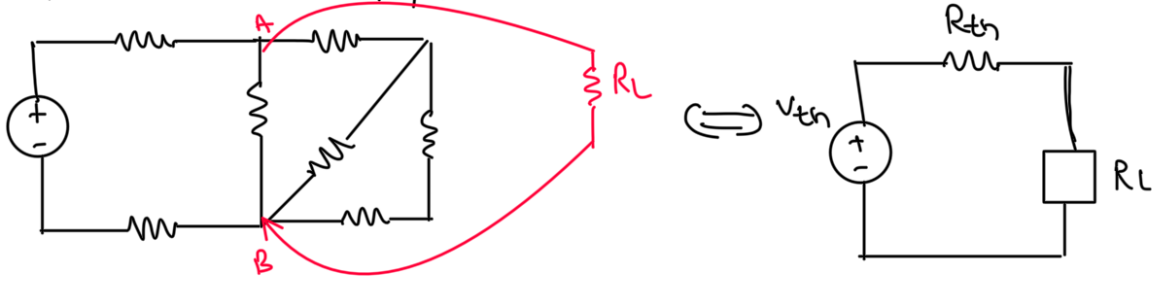
**Superposition Theorm:** Short circuit all voltage sources except in turn. Calculate all the currents in meshes. Current in overall circuit is sum of all those currents. ($i_1 = i_1' + i_1'' + \dots$)

**Superposition Theorm for current sources:**

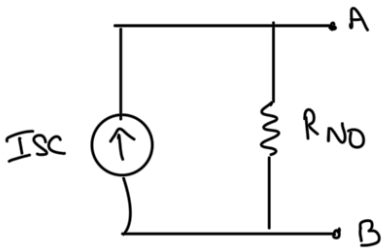
open circuit all but one current source in turn. And the same as above.

General superposition theorm: Take one source and deactivate (short VS, open CS) all the other and solve for current/voltage. Finally add all of them. (only for independent sources)

Thevinin's theorm: Open circuit at $R_L$ (open circuit voltage $V_{OC}$). deactivate independent source. Calculate effective resistance across $R_L$, that is $R_{Th}$.



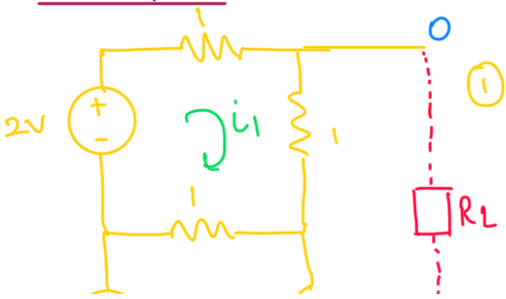

Norton's theorm: $I_{SC}$ is the current through AB when $R_L = 0$ (short circuit current)




$R_{NO}$ is the resistance when seen into AB. (after deactivating all independent sources)

(coincidentaly; $R_{NO} = R_{TH}$

Examples:

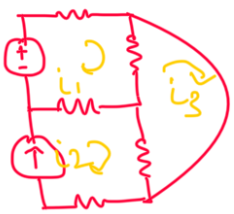


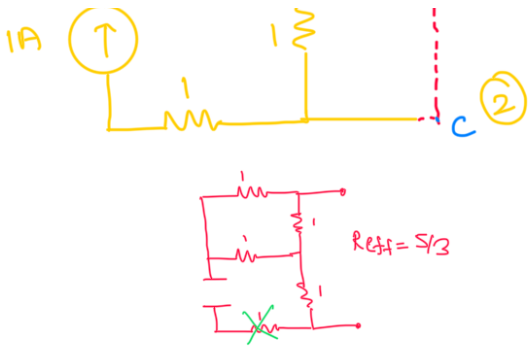

Thevenin

① $-2 + i_1 \cdot 1 + i_1 \cdot 1 + (i_1 - 1)1 = 0$

$\Rightarrow i_1 = 1$

Norton






$i_1 \cdot 1 + 1 \cdot 1 = V_{OC}$

$\Rightarrow V_{OC} = 2$

$R_{TH} = 5/3$




$-2 + i_1 \cdot 1 + (i_1 - i_3) \cdot 1 + (i_1 - i_2) \cdot 1 = 0$

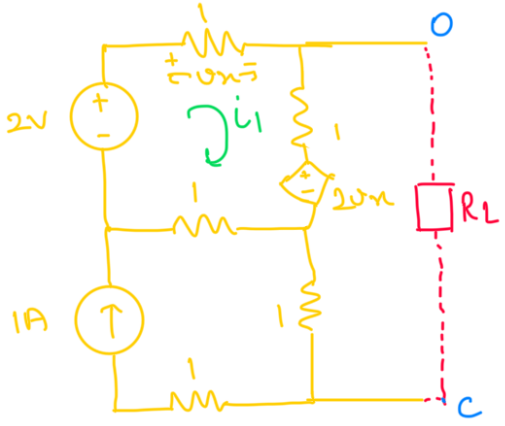
$i_2 + i_3 = 1A$

$(i_1 - i_2) 1 + i_2 \cdot 1 = 0$

$i_3 = I_{SC} = 1 \; ; \; i_2 = 0$

$i_1 = 1$

① $R_{NO} = R_{TH} = 5/3 \Omega$



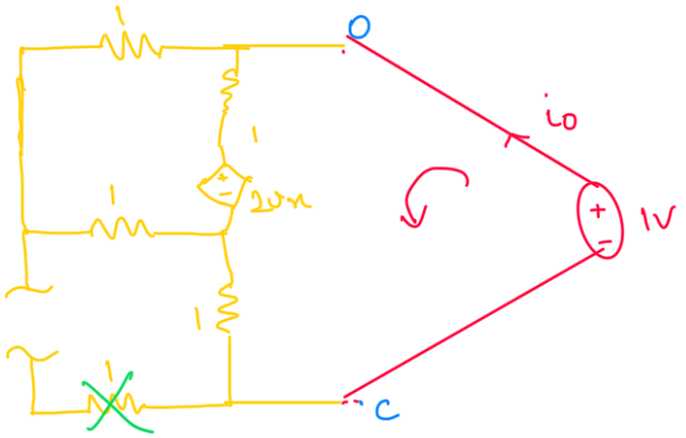

$-2 + i_1 \cdot 1 + i_1 \cdot 1 + 2v_x + (i_1 - 1) \cdot 1 = 0$

($v_x$ = $i_1 \cdot 1$)

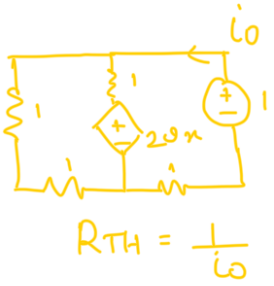
$i_1 = 3/5$

$V_{OC} = 1 \cdot i_1 + 2v_x + 1 \cdot 1 = 14/5$

To find $R_{TH}$




$\Rightarrow$

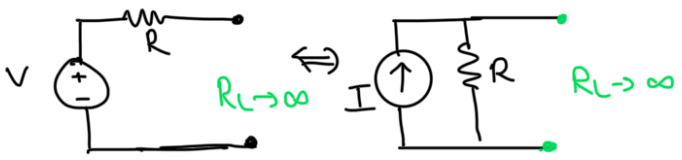



$R_{TH} = \frac{1}{i_o}$

$V_{TH} = I_{SC} \cdot R_{TH}$

we can do one of thevinin or norton (whichever is easier) and convert one to another.

## Source Transformation

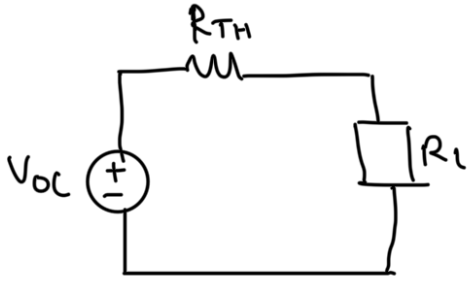



power should match for both when $R_L \to \infty$

$\Rightarrow V = IR$

## Maximum Power Transfer Theorm

Thevenin / Norton → same power drawn as the original circuit




Power at $R_L$ =

$$\Rightarrow \frac{V_{OC}^2}{(R_L + R_{TH})^2} \cdot R_L$$

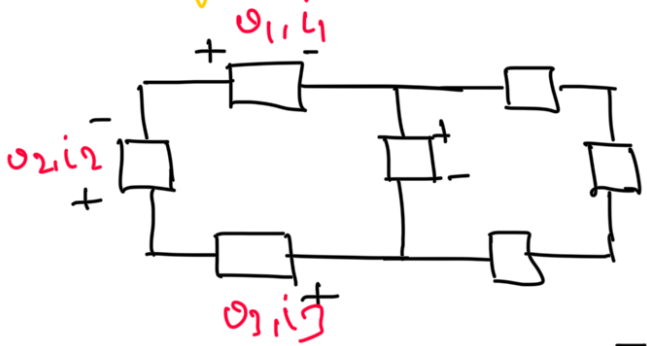
(Maximized when $R_L = R_{TH}$)

$$\boxed{P_{max} = \frac{V_{OC}^2}{4R_{TH}}}$$

$$P_{max} - P_L = \frac{V_{OC}^2}{4R_{TH}} \left[ \frac{R_{TH} - R_L}{R_{TH} + R_L} \right]^2$$

$\Gamma^2 \rightarrow$ reflection coefficient

## Tellegen's Theorm:




For a circuit with any number of elements (non-linear too)

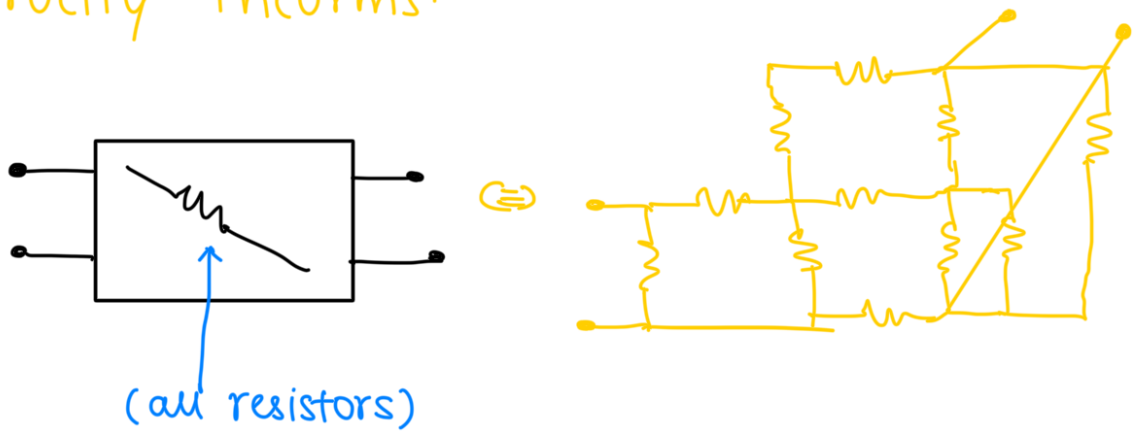
$$\sum_k v_k \cdot i_k = 0$$

Even if $v_k$ and $i_k$ are time dependent,

$$\sum_k v_k(t)\, i_k(t) = 0$$

more general: $\sum v_k(t) i_k(t') = 0$ (for any)

# Reciprocity Theorms:





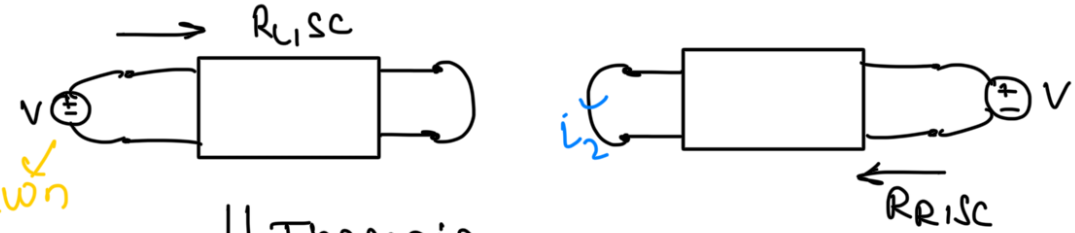


we can do:

current soure ⟷ voltage source

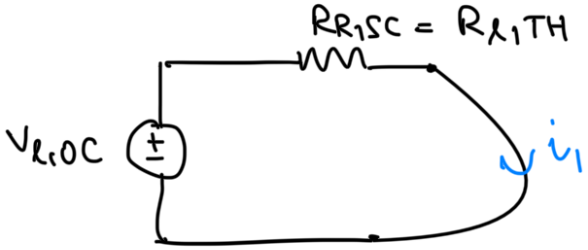
current source ⟺ short circuit

proof:




power drawn
$= V \cdot V / R_{L,SC}$

Thevenin equivalent




$P = V_{L,OC} \times \frac{V_{L,OC}}{R_{R,SC}}$

same thing here

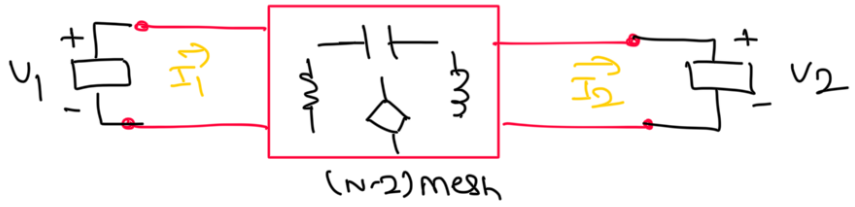
$V_{r,OC} = V \sqrt{\frac{R_{L,SC}}{R_{r,SC}}}$

$i_2 = \frac{V_{r,OC}}{R_{L,SC}} = i_1$

$$\hookrightarrow \frac{V_{1OC}^2}{R_{R1SC}} = \frac{V^2}{R_{R1SC}} \Rightarrow V_{1OC} = V\sqrt{\frac{R_{R1SC}}{R_{R1SC}}}$$

$$i_1 = \frac{V_{1OC}}{R_{R1SC}} = V\sqrt{\frac{1}{R_{R1SC} \times R_{R1SC}}}$$

$\Rightarrow$ both circuits are equivalent




Given the voltages, what are the current

input mesh $I_1(z_{11}) + I_3(z_{13}) + \cdots I_N(z_{1N}) + I_2(z_{12}) = V_1$

output mesh

$$\sum_j z_{1j} = V_1$$

$$\sum_j z_{2j} = -V_2$$

No independent sources $\Rightarrow \sum_j z_{ij} I_j = 0$

$$\begin{bmatrix} z_{11} & z_{12} & z_{13} & \cdots & z_{1N} \\ z_{21} & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ & & & & \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$